

Review_inference_1_top_triples

2026.09.03 - inference_1 is valid, and its top triples are not reproducible across training runs

`inference_2` and `inference_3` are invalid for two compounding reasons, written up in [[010-index-defect|notes-tex/010-index-defect]]. `inference_1` was reported as unaffected by both. This re-derives that rather than taking it on trust, then reads the nominations the run actually makes.

Script: `experiments/010-kuzmin-tmi/scripts/review_inference_1_top_triples.py`. It reads the three stored prediction files under `$DATA_ROOT/data/torchcell/experiments/010-kuzmin-tmi/inference_1/inferred/`, 4,370,595 triples scored by each of the three checkpoints.

Both defect checks pass

Defect two, collapsed triples. Every distinct gene name in the space is resolved through the shared reconciler and checked against the model's gene set. All 526 resolve, none needs alias resolution, so no name was dropped and no triple was silently scored as a double. That is the direct test, since a shortened index list is what produced the 266 collapsed combinations on the panel.

Defect one, the shifted index. A fresh random sample of 3,000 stored triples was re-scored through the validated direct path. Under the 6,607-gene genome map it reproduces the stored predictions at Pearson 0.9999983, maximum absolute difference 3.2e-4, which is the runs' half-precision autocast. The 6,579-gene build map can no longer be tested at all: the guard added after the defect refuses it, raising rather than scoring the wrong genes. The refusal is recorded as the outcome.

So the `inference_1` predictions are the model's answer for the genes they name.

The tail is not reproducible, and this is worse than on the test split

pair	Pearson, all 4.37M	top 10	top 100	top 1,000	top 10,000
M01 vs M02	0.531	0.00	0.14	0.185	0.269
M01 vs M03	0.500	0.00	0.04	0.164	0.193
M02 vs M03	0.544	0.00	0.06	0.248	0.236

No two checkpoints share a single triple in their ten most negative predictions. On the labeled 37,673-record test split the same comparison gives 0.10 to 0.30 at $K = 10$ and 0.39 to 0.47 at $K = 100$, so agreement in the design space is roughly a third of what it is on held-out labels. **Hypothesis (untested):** the gap is a quantile effect, since the top 100 of 4.37M is a far more extreme order statistic than the top 100 of 37,673. Separating that from a genuine architecture instability would need the same K-over-N ratio measured on both.

The practical consequence is that a selection taken from one checkpoint's tail is not the model's opinion. Ranking here is the mean of all three.

The nominations

Top 200 negative by ensemble mean, spanning -0.347 to -0.177 against a training-label standard deviation of 0.0633. All three checkpoints agree on sign for 200 of 200, which is the one strong property of this list. The median spread across the three checkpoints is 0.0694, larger than the label standard deviation itself and a median 33 percent of the nomination's own magnitude, so the ordering within the list carries much less information than the membership of it.

Two structural cautions on the list:

- **One gene carries the tail.** YHL029C appears in 4,402 of the 10,000 most negative triples and in 110 of the top 200. Only 113 distinct genes appear in the top 200 of a 526-gene space.
- **44 of the top 200 contain a gene with zero trigenic training labels.** Every such gene has a trained embedding row, since the graph penalty reached all 6,607 rows, but no trigenic label ever involved it.

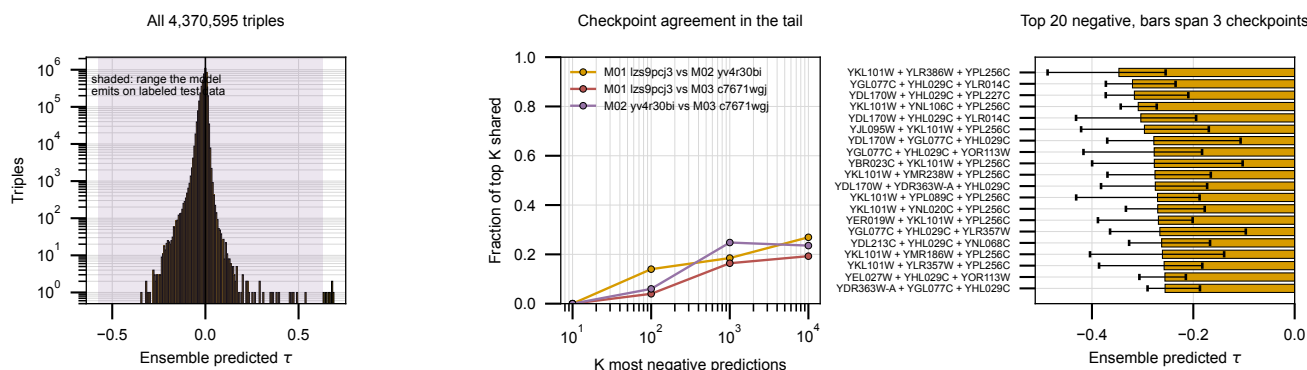
Positives were checked for completeness and behave differently. The top 50 positive predictions run to +0.690, with across-checkpoint spreads of 0.024 to 0.062 and sign agreement on 46 of 50, so they are more reproducible than the negatives. They carry no retrieval validation: the published Kuzmin call is one-sided negative, and the measured retrieval result covers negatives only.

Calibration, and a correction

An earlier reading of this suspected the extreme nominations were outside the range the model produces on labeled data. They are not. Averaged over three checkpoints the test-split predictions span -0.576 to +0.631, against -0.347 to +0.690 here, and only 7 triples of 4,370,595 fall outside the test range. The nominations sit inside the model's demonstrated output range.

What the comparison does show is that the **inference_1** prediction distribution is narrower than the test one, standard deviation 0.0139 against 0.0350. The filters that built this space kept viable, healthy triples, so it holds less of the strong-effect mass the test split contains.

inference_1 trigenic nominations, ensembled over three training runs



What this supports

Reviewing and ensembling the **inference_1** nominations is defensible. Reading a single checkpoint's top ten is not, and neither is treating the within-list ordering as meaningful.

The unresolved question is the same one everything else in 010 is waiting on. These checkpoints trained on a split that is random over records, so a query double seen in training also appears in validation and test. On that split the additive null reaches 0.400 and drops to 0.127 plus or minus 0.033 once query pairs are disjoint. Whether the transformer's margin survives is not measured, in either direction, and **inference_1** combinations whose query double was never screened are exactly the case that measurement would cover.

Outputs

- experiments/010-kuzmin-tmi/results/inference_1_validity.json
- experiments/010-kuzmin-tmi/results/inference_1_checkpoint_agreement.csv
- experiments/010-kuzmin-tmi/results/inference_1_top_triples.csv
- experiments/010-kuzmin-tmi/results/inference_1_gene_frequency.csv